



LFR – Nuevo V2

Team LPPS-1

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1. Introduction

1.1 - What is a Line-Following Robot (LFR)?

A **Line-Following Robot (LFR)** is an autonomous vehicle designed to follow a pre-defined path, usually a black or white line on a contrasting surface. These robots use infrared (IR) sensors to continuously detect the position of the track and adjust their movement accordingly. LFRs are widely used in industrial automation, warehouse logistics, material transportation, and robotics competitions due to their ability to navigate independently with minimal human intervention.

1.2 - About Nuevo V2:

Nuevo V2 is a next gen, high-speed autonomous Line Following Robot (LFR) engineered for maximum efficiency, track stability, and adaptability. Built upon the competitive foundation of its predecessor, Nuevo V2 transitions to an upgraded dual-board architecture, combining a lightweight **25 cm x 15 cm single layer aluminum chassis** with an expanded multi-sensor array for track accuracy. Moving beyond standard **PID (Proportional-Integral-Derivative) control loops**, **Nuevo V2 utilizes an optimized, state-driven logic algorithm paired with a dedicated sub-routine for high-speed intersection detection.** Powered by a high-torque 4-wheel drive TT motor layout and driven by the TB6612FNG dual motor module, the robot eliminates power loss oscillations to achieve rapid acceleration, tight cornering, and consistent execution on university level competition tracks.

1.3 - Wireless Connectivity:

Nuevo V2 includes an onboard **HC-05 Bluetooth module** to handle wireless communication for both testing and broader competition use. Having this hardware built-in removes the need to plug in a USB cable during practice runs, making it much easier to monitor live sensor values, check serial diagnostics, and tune PID values on the fly while the robot is on the track. Additionally, the Bluetooth link allows Nuevo V2 to switch over to **manual Remote Control (RC) mode**. By connecting through a smartphone app, the robot can be driven manually, giving it the flexibility to enter RC-based driving events alongside standard line-following categories. It also leaves room for future features, such as wirelessly adjusting base speeds to match different track surfaces during competitions. Combined with its modular layout and solid drive setup, this makes Nuevo V2 a practical, adaptable platform for testing, competing, and iterating on custom firmware.

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2. Robot's Design

2.1 - Dimensions & Structure:

Nuevo V2 is designed to be compact yet sturdy, ensuring effortless maneuverability and stability at high speeds.

- **Base Material:** Aluminum
- **Length:** 26 cm
- **Width:** 15 cm
- **Height:** 8 cm
- **Weight:** 600g or 1.3 Pounds

The low-profile design keeps the center of gravity low, preventing it from tipping over during sharp turns. The aluminum chassis is lightweight yet strong, providing a solid foundation for motors, sensors, and electronics.

2.2 - Chassis Features:

- Pre-drilled mounting holes for precise placement of motors, sensors, and electronics
 - Even weight distribution to prevent unnecessary skidding
 - Compact form factor for navigating tight corners with ease
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3. Components List

Component	Quantity	Purpose
TB6612FNG Motor Driver	1x	Motor driver
Arduino Nano	1x	Microcontroller
2200mAh 30C 11.7V LiPo	1x	Power Source
Push Button	4x	Used to start and calibrate the LFR algorithm & to navigate the custom OS on OLED.
4 Pin JST Male Header Pins	10x	Allows the different boards to communicate.
4 Pin JST male Connectors	10x	For circuit connections
Vero Board 6x8cm	2x	Base & for soldering connections.
Aluminum RC Car Sheet	1x	Project's Base
TT Motor	4x	DC Motors
TCRT5000 5-Channel Array	1x	IR-Sensors
HC-05	1x	Bluetooth module for communication
LM2596 DC to DC Buck	1x	Voltage Regulator
Mini Buck Convertor	1x	For voltage regulations
Vero Board 3x5cm	1x	For power board.

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4. Wiring and Circuit Connections

Nuevo V2's wiring is kept simple and efficient, ensuring minimal interference while maintaining optimal performance.

4.1 - Arduino Nano to TB6612FNG:

Arduino Nano Pin	TB6612FNG Pin
D5	PWMA
D8	AIN1
D7	AIN2
D12	STBY
D9	BIN1
D10	BIN2
D11	PWMB
5V	VCC
GND	GND

4.2 - TB6612FNG to TT Motors:

TB6612FNG Pin	Motor
A01	Left Motor 1
A02	Left Motor 2
B01	Right Motor 1
B02	Right Motor 2
VM	12V Power Board Output

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4.3 - Arduino Nano to Navigation Buttons:

Arduino Nano	Buttons
D2	OKAY Button
D3	CANCEL Button
D4	UP Button
D6	DOWN Button
GND	Common Ground

4.4 - Arduino Nano to TCRT5000 5-Sensor Array:

Arduino Nano Pin	TCRT5000 Pin
A0	Sensor 1 (L2
A1	Sensor 2 (L1
A2	Sensor 3 (C
A3	Sensor 4 (R1
A6	Sensor 5 (R2
5V	VCC
GND	GND

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4.5 - Arduino Nano to 1.3” OLED

Arduino Nano Pin	OLED
A4	SDA
A5	SCL
5V	5V
GND	GND

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5. Line-Following Algorithm & Control System

Instead of relying on basic logic switching or pure PID control alone, Nuevo V2 combines conditional if/else branching with a PID algorithm for smooth, responsive line tracking.

5.1 - Sensor Data Processing:

The micro-controller continuously polls the front five channel infrared array and calculates a **weighted mean (average position)** from the analog sensor readings. This weighted mean pinpoints the line's exact offset relative to the robot's center line. That error value is fed into a PID calculation loop, which adjusts motor PWM signals on the fly to handle smooth cornering and prevent jerky oscillations. Alongside the continuous PID loop, conditional if/else statements handle state logic and track edge cases:

- **Line Centered (Weighted Mean at Center):** PID output stays minimal, keeping drive motors at maximum forward speed.
 - **Line Drifted (Weighted Mean Left/Right):** PID algorithm applies differential power to the drive wheels to pull the robot back on track without overshooting.
 - **Intersection Detection:** Independent if/else checks monitor the two outer side sensors (A5 & A6) to flag 90° turns or grid intersections, temporarily overriding standard steering to execute sharp maneuvers.
 - **Line Lost (All Sensors Reading White):** An else condition triggers an active recovery sequence, rotating the robot back toward its last known position until the array regains track alignment.
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6. Custom UI

To eliminate reliance on external laptops or serial monitors during competition, Nuevo V2 features an integrated, standalone visual interface powered by a **1.3" I2C OLED display** and driven by a 4-button tactile keypad (UP, DOWN, OKAY, CANCEL). The UI software is built with custom menu-rendering routines that feature smooth transition animations, high-contrast visual indicators, and real-time screen updates, making on-track configuration fast and intuitive.

6.1 - Main Dashboard & Startup Screen:

- **Visual Layout & Animation:** On power-up, the OLED runs a custom splash screen logo animation followed by a smooth fade-in transition to the main system dashboard.
- **Core Functionality:** Displays current robot's model, reboots the built in timer, active mode indication, and quick-access navigation to secondary configuration sub-menus.

6.2 - Live Sensor Monitor & Diagnostics:

- **Visual Layout:** Displays a dynamic visual graph alongside real-time numerical readings. Five vertical bar graphs dynamically fill and drop in real time according to raw IR sensor analog values, with indicator icons flashing when side intersection sensors trigger.
- **Core Functionality:** Allows instant visual confirmation of sensor responses, track contrast, and weighted mean alignment without needing to connect a computer or open a serial plotter.

6.3 - Automated Surface Calibration Mode:

- **Visual Layout:** When selected, a loading spinner animation appears alongside a progress bar that fills dynamically as the robot scans the surface.
- **Core Functionality:** Prompts the user to sweep the sensor array across black and white track sections. The system calculates optimal baseline thresholds automatically and saves them directly to memory, ensuring reliable line detection across varying track surfaces and lighting conditions.

6.4 - Real-Time PID Parameter Tuning:

- **Visual Layout:** Displays a **clean parameter list**, where selecting K_p , K_i , or K_d highlights the row with a inverted-color box. Pressing UP/DOWN increments or decrements values in real time with quick-flash feedback animations.

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- **Core Functionality:** Allows fine-tuning of **Proportional, Integral, and Derivative** gain values right at the track edge. Changes are applied instantly to the active control loop without re-flashing firmware to the Arduino Nano.

6.5 - Drive Mode Selector (LFR & RC Mode):

- **Visual Layout:** Features a dual card toggle animation that smoothly slides between Autonomous LFR Mode and Bluetooth RC Mode.
- **Core Functionality:** Instantly switches control logic between autonomous PID line following and manual wireless driving via the onboard HC-05 Bluetooth module, giving the robot multi-category versatility during competitions.

6.6 - System Impact & Operational Workflow:

By replacing hardcoded parameters and external serial monitoring with an interactive, animated visual UI, Nuevo V2 streamlines the entire workflow from pre-race calibration to live PID parameter adjustments. This self-contained setup drastically cuts setup time between runs, enabling instant troubleshooting and consistent high-speed performance on the track.

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7. Future Upgrades & Applications

7.1 – High-Precision Motion Control & IMU Sensor Fusion:

- **Inertial Measurement Unit Integration:** Integrating a 9-DOF IMU (combining a 3-axis accelerometer, gyroscope, and magnetometer) to measure real-time angular velocity, tilt, and directional heading.
- **Closed-Loop Heading & Inertial Correction:** Using gyroscope and accelerometer data to compensate for wheel slip and drift, enabling precise turns and maintaining straight-line trajectory stability at high speeds.
- **Magnetic Angle Tracking:** Utilizing the magnetometer for long distance directional orientation and grid-based navigation on complex or multi-path competition tracks.
- **Dynamic Speed Management:** Implementing predictive deceleration algorithms that automatically adjust PWM motor outputs based on measured track curvature and angular acceleration during high-speed cornering.

7.2 – Advanced AI Navigation & Path Prediction:

- **Embedded Learning Algorithms:** Implementing lightweight machine learning models directly on board to predict upcoming track profiles based on previous lap telemetry.
 - **Computer Vision Integration:** Combining overhead camera modules with the infrared array to process complex path geometry, obstacles, and multi-lane intersections in real time.
 - **Sensor Fusion Engine:** Merging visual feeds, IR array inputs, and IMU data into a central Kalman filter for accurate track localization under varying lighting and track conditions.
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